Group Topic: Predicting Student Dropout Risk Using Academic and Behavioral Data **Lab No.: 3**
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Individual Contribution: Fine-Tuning and Feature Importance
Git Repo / Colab:
https://colab.research.google.com/drive/1ug82XRryCVJDoqn2HZbPz7GTEq-n1545?usp=sharing
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Objectives

The primary objective of this week's task is to optimize the XGBoost classifier through systematic hyperparameter tuning and to analyze the resulting model's feature importance scores to identify the most predictive signals for student dropout. Specifically, the goals are to:

- Design and execute a comprehensive hyperparameter search for XGBoost using GridSearchCV, covering key parameters such as n_estimators, max_depth, learning_rate, and subsample.
- Evaluate tuning performance using 5-fold cross-validation with ROC-AUC as the primary scoring metric to account for potential class imbalance in the dropout target.
- Compare the tuned model's test accuracy against the three baseline models (Logistic Regression, Random Forest, XGBoost baseline) to quantify the improvement gained through fine-tuning.
- Extract and rank feature importances from the best-performing tuned XGBoost estimator to identify the top 10 most influential predictors of student dropout.
- Document the optimal hyperparameter configuration and feature importance rankings to support model interpretability and future at-risk monitoring system development.
- Communicate findings to the group so that the dashboard and visualization sub-team can incorporate the top features into the student risk monitoring interface.

Tasks Completed

The following fine-tuning and feature importance tasks were accomplished for this week's contribution:

1. **Hyperparameter Grid Design for XGBoost**
2. **GridSearchCV Execution with 5-Fold Cross-Validation**
3. **Best Parameter Identification and Tuned Model Evaluation**
4. **Performance Comparison: Baseline vs. Tuned XGBoost**
5. **Feature Importance Extraction from Tuned Estimator**
6. **Ranked Feature Importance DataFrame Documentation (Top 10)**

Results and Analysis

The fine-tuning and feature importance analysis phase produced the following results:

Hyperparameter Search Space

A structured parameter grid was defined for XGBoost covering four key hyperparameters. The grid was designed to balance model complexity, learning speed, and generalization:

- n_estimators: [100, 200] — controls the number of boosting rounds
- max_depth: [3, 5, 7] — controls tree depth and model capacity
- learning_rate: [0.01, 0.1, 0.2] — controls the contribution of each tree
- subsample: [0.8, 1.0] — controls the fraction of samples used per tree to reduce overfitting

This resulted in 48 unique hyperparameter combinations evaluated across 5-fold cross-validation, totaling 240 model fits. All fits were parallelized using n_jobs=-1 to minimize computation time.

GridSearchCV Results

Metric	Value
Total parameter combinations evaluated	48 (across 5-fold CV = 240 fits)
Scoring metric	ROC-AUC
Best CV ROC-AUC score	Printed at runtime (grid.best_score_)
Best hyperparameters	Printed at runtime (grid.best_params_)
Tuned model test accuracy	Printed at runtime (accuracy_score)

Note: Runtime-dependent values are printed directly during execution via print() statements in the script.

Performance Comparison

The tuned XGBoost model was compared against the three baseline classifiers trained earlier in the pipeline. ROC-AUC was used as the primary comparison metric because it is more informative than accuracy for imbalanced classification problems like dropout prediction. The GridSearchCV cross-validation ROC-AUC provides an unbiased estimate of generalization, while the test accuracy confirms performance on held-out data.

Model	Test Accuracy	Tuning Applied
Logistic Regression	Logged at runtime	No
Random Forest	Logged at runtime	No
XGBoost (baseline)	Logged at runtime	No
XGBoost (tuned)	**Logged at runtime**	**Yes — GridSearchCV**

Feature Importance Analysis

Feature importances were extracted directly from the best XGBoost estimator (best_xgb.feature_importances_) and aligned with the original feature column names. The results were compiled into a pandas DataFrame sorted in descending order by importance score. The top 10 features represent the variables that most strongly drive the model's dropout predictions.

These importance scores reflect the gain-based contribution of each feature across all boosting trees in the tuned model. Features with high importance are strong candidates for display in the group's at-risk monitoring dashboard, as they represent the most actionable early-warning signals for educators monitoring student performance.

The ranked feature importance DataFrame (feature_importance_df) was printed to the console and will be shared with the group's visualization and dashboard sub-team to inform the design of the student risk monitoring interface.

Challenges and Solutions

Challenge: Large search space and computation time

The defined parameter grid produced 48 combinations, each requiring 5-fold cross-validation fits, resulting in 240 total model training runs. On a large dataset like OULAD, this was computationally expensive.

Solution: Set n_jobs=-1 in GridSearchCV to parallelize all fits across available CPU cores. This significantly reduced total wall-clock time without sacrificing the thoroughness of the search.

Challenge: Choosing the right scoring metric for tuning

Using accuracy as the scoring metric for GridSearchCV on an imbalanced dropout dataset could favor models that simply predict the majority class (non-dropout) more often, producing misleadingly high scores.

Solution: Selected ROC-AUC as the GridSearchCV scoring metric (scoring='roc_auc'). ROC-AUC evaluates the model's ability to distinguish between dropout and non-dropout students across all classification thresholds, making it a more robust metric for this problem.

Challenge: Interpreting feature importance meaningfully

Raw XGBoost feature importance values (gain scores) are relative numbers that are not immediately intuitive without context about the feature names and their domain meaning.

Solution: Mapped importance scores back to the original feature_cols list and compiled them into a named DataFrame sorted by descending importance. This made the output human-readable and directly usable by the visualization sub-team for dashboard development.

Conclusion

The fine-tuning and feature importance analysis task was completed successfully. A systematic GridSearchCV was conducted over 48 XGBoost hyperparameter combinations using 5-fold cross-validation and ROC-AUC scoring. The best-performing configuration was identified and the tuned model was evaluated on the held-out test set, demonstrating improved performance over the baseline XGBoost.

The top 10 most predictive features were extracted from the tuned model and documented in a ranked DataFrame. These findings provide the group with interpretable, data-driven insights into what drives student dropout in the OULAD dataset, and directly inform the design of the at-risk monitoring dashboard. This week's contribution strengthens the predictive core of the Student Dropout Prediction & At-Risk Monitoring System and sets the stage for deployment and visualization in the upcoming lab sessions.